

example, how to effectively associate the image features of an abstract definition with its symbolic definition "dangerous" in the same representation space means that we first need to have a deep insight into the intrinsic semantic 'identity' of the very definition between two different types of data, and then discover a reasonable spatial form to reflect this insight. In addition, knowledge alignment based on unified representation must explicitly verify new knowledge's reliability. These updates can not only effectively spread to the specific knowledge conversion process but also, to a certain extent, it is necessary to ensure the consistency of output before and after. However, no matter what, the process should be transparent and explainable. Nevertheless, it is more likely to be closer to solving the current conceptual stability problem of connectionism, as the conceptual structure based on unified representation is constrained by fixed logical rules in formation and update.

Finally, traditional symbolic logic reasoning relies on clearly defined logical rules and logical structures, even if they are represented in real-valued logic, while neural networks reason through fuzzy probability distributions. There are fundamental differences in reasoning mechanisms. Integrating these two reasoning architectures in a unified representation means exploring new reasoning frameworks and developing logical algorithms that simultaneously handle fuzzy and deterministic logic.

5.2 Explainability and Transparency

Neural networks introduce inescapable black-box features and inferences in cooperation with symbolic learning. For the case where they are loosely coupled, the explainability of the neural network cannot get any improvement because even if logical symbols provide rules or constraints for the neural network as embedded vectors, the embedding process itself is not intuitive and requires the addition of complex logical symbol reasoning, and the increased complexity brought about by the interaction between the two. Alternatively, their semantic overlap can form at least partly complementary in explainability when neural networks and symbolic logic utilize the unified representation.

5.3 Sufficient Cooperation

The current integration of neural networks and symbolic logic makes it difficult to avoid the intrinsic problems in both diagrams. For instance, the inexplicable inference, the training cost of neural networks, or the expression limitations and generalization problems of symbolic logic could be introduced to the integrated neural-symbolic model. System complexity and knowledge synchronization may all become new issues. One promising avenue for addressing these challenges is to develop a new model architecture. This architecture would apply an integration layer for the outputs of the neural network component and the symbolic logic component, potentially overcoming the limitations of the current integration. An elastic two-way learning mechanism could be utilized to synchronize their knowledge. However, it is crucial to consider explainability from the outset of the design process.

6 FUTURE RESEARCH DIRECTIONS

6.1 Unified Representations

In Part II of the review, we classify recent neuro-symbolic AI studies into five categories based on information diversity in processing and integrating, as well as the characteristics and capabilities of the representation space in expressing neural network and symbolic logic: unimodal non-heterogeneous, multi-modal non-heterogeneous, single-modal heterogeneous, multi-modal heterogeneous and dynamic adaptive neuro-symbolic AI. This classification method is a good starting point in understanding and characterizing neuro-symbolic AI's performance from the representation

perspective. Therefore, unified representation may be one of the key directions in future breakthroughs, as it minimizes information loss and maximizes knowledge that represents efficiency for both diagrams.

In the meantime, representation space is another promising research direction. Most current neuro-symbolic methods use Euclidean space as the representation space. However, dealing with non-linear problems such as complex relationships, graph-structured data, and timing dependencies is difficult and may not be as efficient as non-Euclidean space. Euclidean space is just a case of non-Euclidean space. Although the latter still needs much exploration regarding specific design methods and combination with Euclidean space, it still provides valuable options for future neuro-symbolic AI research.

6.2 Enhancing Model Explainability

The explainability of models has become an unavoidable challenge in artificial intelligence research. Although the neuro-symbolic AI method provides more substantial transparency than traditional AI to a certain extent, it still cannot meet the requirements for its application in critical fields. Explainability should be considered more during the design phase rather than an afterthought. However, considering the complexity of explainability, we must first establish the basis for explainability. For example, mathematics and physical laws can be regarded as correct standards to a certain extent, while human common-sense logic may be full of contradictions and logical fallacies. Similarly, explainability must be based on a relatively stable concept in neuro-symbolic AI to be more convincing. Therefore, verifying and updating knowledge in LLMs is also an open topic.

The explainability requirements for Neuro-Symbolic AI are essentially divided into two parts: process and result transparency. The former may be based on rigorous logic or formulaic arguments, which means that even if a neural network is used to generate symbols for logical reasoning, this process should be transparent and interpretable enough to verify correctness. The latter shows that some unique thinking habits should also be considered, such as common sense in providing contextual evidence for reasoning results.

Although it is unclear and has obstacles to understanding what is happening in the model, advances in computational neuroscience can still provide some inspiration for future directions. For example, Castañeda et al. [38] found that human deductive reasoning and probabilistic reasoning processes rely on different neurocognitive mechanisms in the brain, and people can suppress prior knowledge for deductive reasoning according to task requirements. Not all reasoning processes can be attributed to probability. These insights may imply that our prior knowledge of deductive reasoning should be tailored to the specific situation. Trumpp et al. [180] found that activity in sensory and motor areas during conceptual processing can also occur unconsciously. Popp et al. [146] also found that words related to actions and sounds activate sensory and motor areas of the brain during conceptual processing. This result shows that the acquisition of conceptual knowledge relies on reproducing sensory and motor brain networks, which supports the view of ground cognitive theory. Designing neuro-symbolic AI with a stable working or knowledge memory structure may inspire brain function. Belekou et al. [26] found significant differences in brain activation patterns when processing paradox reasoning and deductive reasoning tasks. Coetzee et al. [51] found that logical reasoning in the adult brain may be separated from language processing. This research may inspire us to consider more flexible reasoning paths in neuro-symbolic AI or even dynamically configurable task reasoning methods.

6.3 Ethical Considerations and Social Impact

Ethical and social considerations are another inescapable problem. If a high portion of our content will come from generative AI in the future, then the significance of this content will be far beyond the scope of being measured by

credibility alone. All moral requirements of today’s society, such as fairness and justice, privacy protection, prejudice and discrimination, environmental ethics, technological ethics, humanitarianism, or even religion, should be included in the evaluation criteria of AI algorithms. Just as the research by [184] shows, the significant correlation between computational models and the human brain does not prove that they have been implemented. As with the same process, indirectly interpreting the output of black-box AI does not meet process transparency requirements, so we may need designs that can fundamentally explain the generation process. Some research points out that neural-symbolic AI can provide more significant help in this regard by providing more rules and constraints for processing neural networks, such as [27] proposed the application of the BDI(belief-desire-intention) model and reinforcement learning method to simulate how subjects internalize norms and make decisions consistent with human moral judgment, providing effective modeling and examples in Neuro-Symbolic AI in the future. It provides a possible inspiration for complex moral behavior.

In summary, we discussed the possible future development directions of neuro-symbolic AI from three aspects: unified representation, enhancing model explainability, ethical considerations, and social impact. These three aspects represent expectations for Neuro-Symbolic AI at three different levels. The representation space determines the technical foundation and the extent to which we can achieve reasonable explainability and transparency requirements, affecting the overall application prospects of neuro-symbolic AI and its impact on ethics and society.

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